

高阶偏导数.

若 $u=f(x,y)$ 关于 x, y 有偏导数, 若 $f_x(x,y), f_y(x,y)$ 还有偏导数: $(f_x(x,y))_x, (f_x(x,y))_y, (f_y(x,y))_x, (f_y(x,y))_y$ 称为二阶偏导数

$$\left. \begin{aligned} (f_x(x,y))_y &= f_{xy}(x,y) \\ (f_y(x,y))_x &= f_{yx}(x,y) \end{aligned} \right\} \text{混合偏导数}$$

$$\text{例 } f(x,y) = \begin{cases} xy \frac{x^2-y^2}{x^2+y^2}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases}$$

$$\text{在 } (0,0): \begin{matrix} f_{xy}(0,0) \\ f_{yx}(0,0) \end{matrix}$$

$$f(x,0) = \begin{cases} xy \cdot \frac{x^2-y^2}{x^2+y^2} \Big|_{y=0} & x \neq 0 \\ 0 & x = 0 \end{cases} = \begin{cases} 0, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

$$\text{在 } (0,0): f_x(0,0) = \frac{d}{dx} f(x,0) \Big|_{x=0} = 0;$$

$$\text{同理: } f_y(0,0) = 0,$$

$$y \neq 0: f(x,y) = xy \frac{x^2-y^2}{x^2+y^2} :$$

$$\begin{aligned} f_x(x,y) &= y \frac{(x^2-y^2)'(x^2+y^2) - (x^2-y^2) \cdot 2x}{(x^2+y^2)^2} \\ &= \frac{y(x^4-y^4) + 2x^2y(x^2+y^2) - 2x^2y(x^2-y^2)}{(x^2+y^2)^2} \\ &= \frac{y(x^4-y^4) + 4x^2y^3}{(x^2+y^2)^2} \end{aligned}$$

$$f_y(x,y) = - \frac{x(y^4-x^4) + 4x^3y^2}{(x^2+y^2)^2}.$$

$$\left. \begin{array}{l} y \neq 0 \text{ 时: } f_x(0, y) = -y \\ y = 0: f_x(0, 0) = 0 \end{array} \right\} \begin{array}{l} \forall y \in \mathbb{R}: \\ f_x(0, y) = -y \Rightarrow f_x(0, 0) = -1. \end{array}$$

$$\left. \begin{array}{l} \text{同理: } x \neq 0: f_y(x, 0) = x \\ x = 0: f_y(0, 0) = 0 \end{array} \right\} \begin{array}{l} \forall x \in \mathbb{R}: \\ f_y(x, 0) = x: f_y(0, 0) = 1 \end{array}$$

$$\text{在 } (0, 0): f_{xy}(0, 0) \neq f_{yx}(0, 0).$$

原因: $f_{xy}(x, y), f_{yx}(0, 0)$: 不连续.

定理 1.6. $f_{xy}(x, y), f_{yx}(x, y)$ 连续, 则相等.

注: $D \subset \mathbb{R}^2$ 区域
① 已知: $u = f(x, y)$ 有连续偏导数 $\Rightarrow$ 可微

若 $f_x(x, y), f_y(x, y)$ 连续, 则 $f(x, y)$ 连续可微

$$du = f_x(x, y) \Delta x + f_y(x, y) \Delta y \quad \Delta x, \Delta y \text{ 为独立变量}$$

$\Rightarrow$ 关于 (x, y) 连续.

$$f(x, y) \in C^1(D) \Leftrightarrow f_x(x, y), f_y(x, y) \text{ 在 } D \text{ 中连续}$$

$$C^1(D) = \{f(x, y) \mid f(x, y) \text{ 在 } D \text{ 中连续可微}\}.$$

$$\textcircled{2} \text{ 若 } u = f(x, y), f_{xx}(x, y), f_{xy}(x, y), f_{yx}(x, y), f_{yy}(x, y)$$

在 D 中连续, 则称 $f(x, y)$ 为二阶连续可微,

$$C^2(D) = \{f(x, y) \mid f(x, y) \text{ 在 } D \text{ 中二阶连续可微}\},$$

$m \geq 1$: $u = f(x, y)$ 在 D 中有 m 阶连续偏导数

称 $f(x, y)$ 在 D 中 m 阶连续可微

$$C^m(D) = \{f(x, y) \mid f(x, y) \text{ 在 } D \text{ 中 } m \text{ 阶连续可微}\}.$$

三. 微分全微分:

$u = f(x, y)$ 在区域 $E \subset \mathbb{R}^2$ 中: 可微. 其微分

$$du = f_x(x, y) \Delta x + f_y(x, y) \Delta y.$$

$\Delta x, \Delta y$ 为自变量. 固定: du 是 $(x, y) \in D$ 之函数

若 du 可微:

$$d(du) = \partial_x(du) \Delta x + \partial_y(du) \Delta y$$

称 $u = f(x, y) \Rightarrow$ 二阶全微分: 记 d^2u

$$\begin{aligned} d^2u &= d(du) = \partial_x(du) \Delta x + \partial_y(du) \Delta y \\ &= (f_x(x, y) \Delta x + f_y(x, y) \Delta y)_x \Delta x \\ &\quad + (f_x(x, y) \Delta x + f_y(x, y) \Delta y)_y \Delta y \\ &= (f_{xx}(x, y) \Delta x + f_{yx}(x, y) \Delta y) \Delta x \\ &\quad + (f_{xy}(x, y) \Delta x + f_{yy}(x, y) \Delta y) \Delta y \end{aligned}$$

$$\frac{f_{xy}(x, y) = f_{yx}(x, y)}{f(x, y) = x: \Rightarrow dx = \Delta x}$$

$$= f_{xx} (\Delta x)^2 + 2 f_{xy} \Delta x \Delta y + f_{yy} (\Delta y)^2$$

$$f(x, y) \equiv y \Rightarrow dy = \Delta y, \quad dx^2 = dx \cdot dx \quad dy^2 = dy \cdot dy$$

$$\text{例5: } du = f_x(x, y) dx + f_y(x, y) dy$$

$$d^2u = f_{xx}(x, y) dx^2 + 2f_{xy}(x, y) dx dy + f_{yy}(x, y) dy^2$$

四. 一元多元函数的全微分:

$$u = f(x), \quad x = (x_1, x_2, \dots, x_n) \in E \subset \mathbb{R}^n.$$

$$x \rightarrow x + \Delta x, \quad \Delta x = (\Delta x_1, \Delta x_2, \dots, \Delta x_n) \in \mathbb{R}^n.$$

$$\text{若 } \Delta u = f(x + \Delta x) - f(x) \quad \rho = |\Delta x| = \sqrt{\sum_{i=1}^n (\Delta x_i)^2}$$

$$= A_1 \Delta x_1 + A_2 \Delta x_2 + \dots + A_n \Delta x_n + o(\rho)$$

若 $u = f(x)$ 在 x 可微, 则 $A_1 \Delta x_1 + \dots + A_n \Delta x_n \xrightarrow{\rho \rightarrow 0} u - f(x)$

在微分中, 记为

$$du = A_1 \Delta x_1 + A_2 \Delta x_2 + \dots + A_n \Delta x_n$$

$$\Rightarrow A_i = \frac{\partial f(x)}{\partial x_i}$$

$$du = \sum_{i=1}^n \frac{\partial f(x)}{\partial x_i} dx_i$$

在近似计算中:

$y = f(x)$: 在 x_0 邻域中 近似可微.

$$\Delta x: \text{"d."} \quad \rho = |\Delta x| \text{"d."} \quad x = x_0 + \Delta x$$

$$f(x) - f(x_0) = f(x_0 + \Delta x) - f(x_0) = \sum_{i=1}^n f_{x_i}(x_0) \Delta x_i + \underline{o(\rho)}$$

$$\Rightarrow f(x) \approx f(x_0) + \sum_{i=1}^n f_{x_i}(x_0) \Delta x_i$$

例 求: $1.03^{1.98}$

设: $f(x, y) = x^y$; $x_0 = 1, y_0 = 2,$

$$\Delta x = 1.03 - x_0 = 0.03, \Delta y = 1.98 - y_0 = -0.02$$

$f(x, y)$ 在 $(1, 2)$ 处可微.

$$f_x(x, y) = y \cdot x^{y-1}; \quad f_y(x, y) = x^y \cdot \ln x$$

$$1.03^{1.98} = f(1.03, 1.98)$$

$$\approx f(1, 2) + f_x(1, 2) \cdot \Delta x + f_y(1, 2) \Delta y$$

$$\begin{aligned} x_0 = 1, y_0 = 2: \quad &= 1^2 + \underbrace{y_0 \cdot x_0^{y_0-1}}_{\downarrow} \cdot \Delta x + \underbrace{x_0^{y_0} \ln x_0}_{\text{!}} \cdot \Delta y \\ &= 1 + 2 \cdot 1^{2-1} \cdot 0.03 + \ln 1 = 0 \\ &= 1.06 \quad \# \end{aligned}$$

§2 复合变数与偏导数

二元函数 $z = f(x, y)$ 与 $\begin{cases} x = g(u, v) \\ y = h(u, v) \end{cases}$ 复合, 得到

关于 u, v 的二元函数:

$$z = f(g(u, v), h(u, v))$$

定理 2.1 设 $z = f(x, y)$ 在 $M_0(x_0, y_0)$ 邻域 $O(M_0, r)$ 中可微, $x = g(u, v), y = h(u, v)$ 在 $P_0(u_0, v_0)$ 邻域 $O(P_0, \rho)$

中可微, $(g(p_0), h(p_0)) = M_0(x_0, y_0)$ 且 $g(p_0) = x_0, h(p_0) = y_0$
 $(g(u, v), h(u, v)) \in O(M_0, r)$.

则 $z = f(\underline{g(u, v)}, \underline{h(u, v)})$ 在 $O(p_0, a)$ 中关于 u, v 可微且

$$\frac{\partial z}{\partial u} = f_x \cdot g_u + f_y \cdot h_u,$$

$$\frac{\partial z}{\partial v} = f_x \cdot g_v + f_y \cdot h_v.$$

证: $z = f(x, y) = f(g(u, v), h(u, v))$.

对 u, v 附近 $(\frac{u_0}{2}, \frac{v_0}{2})$ 取 $\Delta u, \Delta v$:

$$\Delta z = f(\underline{g(u+\Delta u, v+\Delta v)}, \underline{h(u+\Delta u, v+\Delta v)}) - f(g(u, v), h(u, v)).$$

$$\begin{aligned} \Delta x &= \underline{g(u+\Delta u, v+\Delta v)} - g(u, v) & g(u, v) \text{ 可微,} \\ &= g_u(u, v) \Delta u + g_v(u, v) \Delta v + o_1(\rho) & \rho = \sqrt{(\Delta u)^2 + (\Delta v)^2} \end{aligned}$$

$$\begin{aligned} g(u+\Delta u, v+\Delta v) &= g(u, v) + \Delta x \\ &= \underline{x} + \Delta x & x = g(u, v) \end{aligned}$$

$$\begin{aligned} \Delta y &= h(u+\Delta u, v+\Delta v) - h(u, v) & h(u, v) \text{ 可微} \\ &= h_u(u, v) \Delta u + h_v(u, v) \Delta v + o_2(\rho) \end{aligned}$$

$o_1(\rho), o_2(\rho)$: $\frac{o_i(\rho)}{\rho} \rightarrow 0$ 当 $\rho \rightarrow 0$ 时

$$\begin{aligned} h(u+\Delta u, v+\Delta v) &= h(u, v) + \Delta y \\ &= y + \Delta y \end{aligned}$$

$\rho \rightarrow 0$ 时: $\Delta x \rightarrow 0 \quad \Delta y \rightarrow 0$.

$$\begin{aligned}\Delta z &= f(g(u+\Delta u, v+\Delta v), h(u+\Delta u, v+\Delta v)) - f(g(u, v), h(u, v)) \\ &= f(x+\Delta x, y+\Delta y) - f(x, y) \quad \downarrow \quad \underline{z=f(x, y) \text{ 可微}} \\ &= \underline{f_x(x, y) \Delta x} + f_y(x, y) \Delta y + o(\rho_1)\end{aligned}$$

$$\rho_1 = \sqrt{(\Delta x)^2 + (\Delta y)^2} \rightarrow 0$$

$$= f_x(x, y) \left[g_u(u, v) \Delta u + g_v(u, v) \Delta v + o_1(\rho) \right]$$

$$+ f_y(x, y) \left[h_u(u, v) \Delta u + h_v(u, v) \Delta v + o_2(\rho) \right] + o(\rho_1)$$

$$\Delta z \neq A \Delta u + B \Delta v + o(\rho) \quad \rho = \sqrt{(\Delta u)^2 + (\Delta v)^2}$$



$$A, B \in \mathbb{R}$$

$$\frac{o(\rho)}{\rho} \rightarrow 0 \quad (\rho \rightarrow 0)$$

$$z = f(g(u, v), h(u, v)) : \text{关于 } u, v \text{ 可微}$$

$$\Delta z = (f_x \cdot g_u + f_y \cdot h_u) \Delta u + (f_x \cdot g_v + f_y \cdot h_v) \Delta v$$

$$+ \underbrace{f_x \cdot o_1(\rho)}_{\text{red circle}} + \underbrace{f_y \cdot o_2(\rho)}_{\text{red circle}} + o(\rho_1).$$

→ 可微
不可微

$$\frac{f_x}{\text{有界}}$$

$$\frac{o_1(\rho)}{\rho} \xrightarrow{\rho \rightarrow 0} 0;$$

$$\frac{f_y \cdot o_2(\rho)}{\rho} \xrightarrow{\rho \rightarrow 0} 0$$

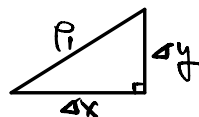
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可微

$$\frac{o(\rho)}{\rho} = \underbrace{\frac{o(\rho)}{\rho_1}}_{\downarrow 0} \cdot \underbrace{\frac{\rho_1}{\rho}}_{\leftarrow \text{有界}}$$

$$\rho = \sqrt{(\Delta u)^2 + (\Delta v)^2} \rightarrow 0 \text{ 时}$$

$$\text{有 } \rho_1 = \sqrt{(\Delta x)^2 + (\Delta y)^2} \rightarrow 0;$$



$$\frac{\rho_1}{\rho} = \frac{\sqrt{(\Delta x)^2 + (\Delta y)^2}}{\rho} \leq \frac{|\Delta x|}{\rho} + \frac{|\Delta y|}{\rho}$$

$$\begin{aligned} \frac{|\Delta x|}{\rho} &= \left| \frac{g_u \Delta u + g_v \Delta v + o_1(\rho)}{\rho} \right| \\ &\leq \underbrace{|g_u| \frac{|\Delta u|}{\rho} + |g_v| \frac{|\Delta v|}{\rho}}_{\text{有界}} + \underbrace{\left| \frac{o_1(\rho)}{\rho} \right|}_{\text{有界} \rightarrow 0} \quad (\rho \rightarrow 0) \end{aligned}$$

$$\begin{aligned} \therefore \left| \frac{\Delta u}{\rho} \right| &\leq 1 \\ \left| \frac{\Delta v}{\rho} \right| &\leq 1 \end{aligned}$$

同理 $\frac{|\Delta y|}{\rho}$ 有界

(极限存在, 局部有界)

$\therefore 0 \leq \frac{\rho_1}{\rho}$ 有界.

$$\Rightarrow \frac{o(\rho)}{\rho} \xrightarrow{\rho \rightarrow 0} 0$$

$$\begin{aligned} \therefore \Delta z &= (f_x \cdot g_u + f_y \cdot h_u) \Delta u \\ &\quad + (f_x \cdot g_v + f_y \cdot h_v) \Delta v + o(\rho). \end{aligned}$$

$$\frac{o(\rho)}{\rho} \rightarrow 0 \quad (\rho \rightarrow 0), \quad \rho = \sqrt{(\Delta u)^2 + (\Delta v)^2} \rightarrow 0$$

$\therefore z = f(g(u, v), h(u, v))$ 关于 u, v 可微,

$$A \quad \frac{\partial z}{\partial u} = f_x \cdot g_u + f_y \cdot h_u,$$

$$\frac{\partial z}{\partial v} = f_x \cdot g_v + f_y \cdot h_v. \quad \#$$

$$z = f(x, y) \begin{array}{l} \xrightarrow{\frac{\partial z}{\partial u}} f_x \cdot x_u + f_y \cdot y_u \\ \xrightarrow{\frac{\partial z}{\partial v}} f_x \cdot x_v + f_y \cdot y_v. \end{array}$$

例 $z = x^2 e^y$, $x = 4u$, $y = 3u^2 - 2v$: 求 z_u, z_v .

解 ① $z = (4u)^2 e^{3u^2 - 2v}$; $\Rightarrow z_u, z_v$;

$$\begin{aligned} \text{② } \frac{\partial z}{\partial u} &= z_x \cdot x_u + z_y \cdot y_u \\ &= 2x e^y \cdot 4 + x^2 e^y \cdot 6u \\ &= 8x e^y + 6x^2 u e^y \leftarrow x, y \text{ 不变} \end{aligned}$$

z_v : 类似.

例 $z = f(xy, \frac{x}{y})$, 求 z_x, z_{xx}, z_{xy}

解: 记 $u = xy$, $v = \frac{x}{y}$: $z = f(u, v)$

$z = f(u(x, y), v(x, y)).$

$$\frac{\partial z}{\partial x} = f_u \cdot \underline{u_x} + f_v \cdot v_x = f_u(u, v) \cdot y + f_v(u, v) \cdot \frac{1}{y}.$$

$$\begin{aligned}
\frac{\partial^2 z}{\partial x^2} &= f_{uu} \cdot (u_x) \cdot y + f_{uv} \cdot v_x \cdot y \\
&\quad + \underline{f_{vu} \cdot (u_x) \cdot \frac{1}{y}} + \underline{f_{vv} \cdot v_x \cdot \frac{1}{y}} \\
&= f_{uu} \cdot y^2 + f_{uv} \cdot \frac{y}{y} \cdot y + f_{vu} \cdot y \cdot \frac{1}{y} + f_{vv} \cdot \frac{1}{y} \cdot y \\
&= f_{uu} \cdot y^2 + \underline{2f_{uv}} + f_{vv} \cdot \frac{1}{y^2}.
\end{aligned}$$

$$\begin{aligned}
\frac{\partial^2 z}{\partial y \partial x} &= \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial x} \right) = \frac{\partial}{\partial y} \left(f_u(u,v) \cdot y + f_v(u,v) \cdot \left(\frac{1}{y} \right) \right) \\
&= f_{uu} \cdot u_y \cdot y + f_{uv} \cdot v_y \cdot y + f_u \cdot 1 \\
&\quad + \underline{f_{vu} \cdot u_y \cdot \frac{1}{y}} + f_{vv} \cdot v_y \cdot \frac{1}{y} + f_v \cdot \left(-\frac{1}{y^2} \right) \\
&= f_{uu} \cdot x \cdot \frac{1}{y} + f_{uv} \cdot \left(-\frac{x}{y^2} \right) + f_u \\
&\quad + f_{vu} \cdot x \cdot \frac{1}{y} + f_{vv} \cdot \left(-\frac{x}{y^2} \right) \cdot \frac{1}{y} - \frac{1}{y^2} f_v \\
&= \dots
\end{aligned}$$

练习 练习. #

作业: P118. 1, 4, 6, 7, 9